

The empirical recurrence for OEIS A237220: recovering a conjecture that is not written down

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2 September 2026

Abstract

OEIS A237220 records “Empirical recurrence of order 88”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 220$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 88, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 88 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A237220 is “Number of $(n+1) \times (2+1)$ 0..3 arrays with the maximum plus the lower median minus the upper median of every 2×2 subblock equal.”. It has offset 1 and begins

1416, 14472, 161598, 2006382, 25423162, 338131394, 4474548444, 60750922148, ...

The entry states:

Empirical recurrence of order 88 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 09:59:04, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 220$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 220$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 440$ exact terms of the model returns order 88 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{88} c_i a(n-i),$$

c_1	42	c_{23}	−137673459136688794336	c_{45}	−13971894386700081780288560288
c_2	−360	c_{24}	−292017139130458019369	c_{46}	−32863987913598556276907983923
c_3	−8796	c_{25}	1980651903286791709362	c_{47}	41739162709343243099989995173
c_4	180605	c_{26}	1299245441703267675837	c_{48}	57777673904334430178977267964
c_5	−85611	c_{27}	−21529396739842927507284	c_{49}	−98003686228575705222324307847
c_6	−23295356	c_{28}	6360200787061581581383	c_{50}	−80328726536637592492365874817
c_7	150129267	c_{29}	182651332251338883429904	c_{51}	186503235497069257081320046246
c_8	1262431138	c_{30}	−190547205234313147711382	c_{52}	82588794549792606431341433382
c_9	−16746759501	c_{31}	−1223956295952180121000054	c_{53}	−291483437492615791313528636966
c_{10}	−12475892762	c_{32}	2139256025612289286676280	c_{54}	−49010366432070721646037296582
c_{11}	947714154897	c_{33}	6454331086355185910101923	c_{55}	375951930428547490463456702126
c_{12}	−2408523973938	c_{34}	−16526462249884242394567714	c_{56}	−17338520741304150703855236538
c_{13}	−32168478400458	c_{35}	−26073792358656427777260226	c_{57}	−400023549481002077316084240184
c_{14}	170367973427493	c_{36}	98100635888331479945043228	c_{58}	89518705210186742027446547592
c_{15}	640523629353028	c_{37}	73741079053825581386103117	c_{59}	349532114656044357181926770444
c_{16}	−6282156783967400	c_{38}	−466166500862436146682271709	c_{60}	−133169097448783132062071397064
c_{17}	−4290846255146685	c_{39}	−90000222803419039529109662	c_{61}	−248566511897314238454935026416
c_{18}	154117743524753231	c_{40}	1807349527194395185945502890	c_{62}	132116634746193611761896832896
c_{19}	−164623437237695244	c_{41}	−417675277965351684054917698	c_{63}	141759231326461300261471355824
c_{20}	−2691846689195909751	c_{42}	−5767766606228172415089862379	c_{64}	−98012970969858368239679602304
c_{21}	6665761134347222518	c_{43}	3388573953017476930557968185	c_{65}	−63289401770139019994520423392
c_{22}	33862690421687102075	c_{44}	15185747126057263567991362811	c_{66}	56148040125805875618159121472

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 88, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $88 + 4$ terms removed returns order 88 as well, so $\deg q_{\min} = 88 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once

more on the sequence with its first $88 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 88 that this sequence satisfies — and that family turns out to have exactly one member.

References

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